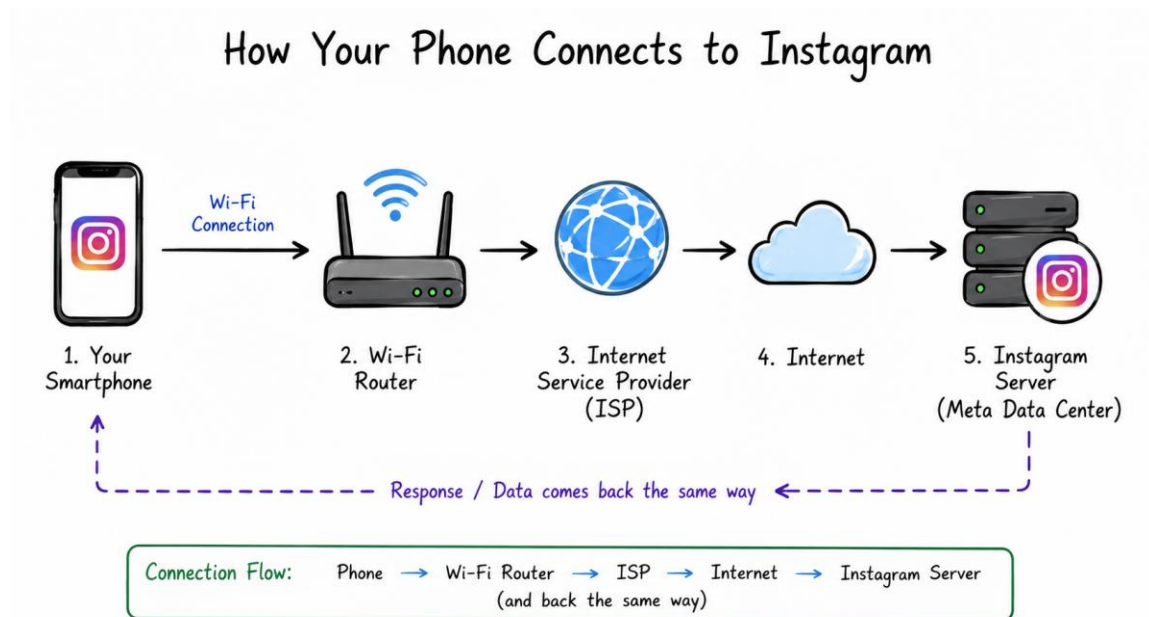


# Introduction to Networks

- 1) Draw a simple diagram (hand-drawn or digital) showing how your phone connects to Instagram, including your device, WiFi router, ISP, and Instagram server. Label each part and the connections.



- 2) List 5 devices you use or see daily that are part of a network (for example: smartphone, smart TV, WiFi router, laptop, smartwatch). For each, write what type of network it connects to (LAN, WAN, PAN, etc.).

| Device       | Network Type                    | Purpose                               |
|--------------|---------------------------------|---------------------------------------|
| Smartphone   | WAN (Mobile Data) / LAN (Wi-Fi) | Internet browsing and apps            |
| Laptop       | LAN (Wi-Fi)                     | Access internet and local network     |
| Smart TV     | LAN                             | Streaming videos from the internet    |
| Smartwatch   | PAN (Bluetooth)                 | Connects to smartphone                |
| Wi-Fi Router | LAN & WAN                       | Connects home devices to the internet |

- 3) Explain the difference between the Internet and an intranet using an example from your college or a public WiFi you have used.

| Internet                             | Intranet                                                    |
|--------------------------------------|-------------------------------------------------------------|
| Public network available worldwide.  | Private network used within an organization.                |
| Anyone can access it.                | Only authorized users can access it.                        |
| Example: Google, Instagram, YouTube. | Example: College student portal or company employee portal. |

**Example:**

- **Internet:** I use my college Wi-Fi to access YouTube and Instagram.
- **Intranet:** My college's student portal for attendance, results, and notices is only available to students and staff.

4) Pick any one app you use (like WhatsApp, Zomato, or Flipkart) and describe, in 4-5 lines, how data travels from your device to their server and back when you perform a search inside the app.<br><br><em><strong>Hint:</strong> Mention the devices and networks involved, and what happens when you hit 'search'.</em>

- When I search for a contact or message in WhatsApp, my smartphone sends the search request through the **Wi-Fi router** or **mobile network** to my **ISP**. The ISP forwards the request through the **Internet** to the **WhatsApp server**. The server processes the request and sends the matching results back through the Internet, ISP, and router to my smartphone. When I press the **Search** button, the app quickly communicates with the server and displays the requested information.